GGG-F-671b
JULY 11, 1957
SUPERSEDING
Fed. Spec. GGG-F-671a
April 8, 1954

FEDERAL SPECIFICATION

FRAME, HAND HACKSAW; FRAME, JEWELER'S SAW; AND FRAME, COPING SAW

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers saw frames used in conjunction with blades for manually cutting materials of different composition such as metal, wood, and plastic. Frames furnished under this specification shall be without blades.

1.1.1 *Federal specification coverage.*—Federal specifications do not cover all types and sizes of the commodity indicated by the title of the specification, but are intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, styles, and grades.*—Frames shall be of the following types, classes, styles and grades, as specified (see 6.2):

Type I.—Hack, adjustable length.
- Class 1.—Flat frame, open grip.
  - Style A.—Shallow throat.
    - Grade a.
    - Grade b.
  - Style B.—Medium throat.
- Class 2.—Flat frame, closed grip.
- Class 3.—Round frame, closed grip.

Type II.—Hack, adjustable throat.

Type III.—Hack, nonadjustable.
- Class 1.—General purpose.
- Class 3.—Deep throat.

Type IV.—Hack, close quarters.
- Class 1.—Straight frame.
- Class 2.—Triangular frame.

Type V.—Jeweler's, adjustable length.
- Style A.—Medium throat.
- Style B.—Deep throat.

Type VI.—Jeweler's, nonadjustable.
- Style A.—Medium throat.
- Style B.—Deep throat.

Type VII.—Coping.
- Style A.—Medium throat.
- Style B.—Deep throat.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-P-31 — Preservation, Packaging, Packing, and Marking of Material; and Shipping Instructions (Domestic Shipment and Storage).

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Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–P–15424 — Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. — The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Governmental:

National Bureau of Standards Handbook:

H28 — Screw-Thread Standards for Federal Services.

(Copies of the National Bureau of Standards Handbook may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

Nongovernmental:

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules and Container Specifications.

(Copies may be obtained from the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Illustrations.—The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of frames otherwise conforming to this specification.

3.2 Material.—Saw frames shall be of high-quality steel or of material specified herein, and the heat treatment employed shall be such as to produce frames conforming to the requirements specified herein.

3.3 Design and construction.—The frames shall be adjustable or nonadjustable, as specified herein, and so constructed as to hold the blade securely under high tensions and maintain rigidity during hard, long use. All frames, except type II, shall be so designed that all parts will remain assembled when blades are not in place.

3.3.1 *Handle grips.* — Handles shall be shaped to provide a comfortable handgrip. Composition-material handle grips, such as

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plastic, shall withstand the tests specified in 4.4.3.

3.3.2 *Blade-tension adjustment.* — All *frames, except type* IV, class 2, and type V, style B, shall be provided with a hand adjustment for regulating and producing high blade tension, and shall maintain the initial tension adjustment.

3.3.2.1 *Handle-tension style.* — In the handle-tension style of frame the threaded sleeve within the handle shall be adequately secured to prevent it from turning or being withdrawn from the handle under high tension. A friction washer shall be provided between the handle and the frame. The length of the threaded section of the sleeve shall be not less than one and one-half times the nominal diameter of the threaded section of the blade-holding device, except that for jeweler's and coping saw frames, the minimum length shall be four-fifths of the nominal diameter of the threaded section.

3.3.2.2 *Thumb-nut-tension style.* — In the thumb-nut-tension style of frame the blade holder(s) shall pass through an opening in the frame or a sleeve which shall be rigidly secured to the frame. One holder shall be threaded to accommodate a thumb nut. The threaded section of the thumb-nut shall be not less than four-fifths of the nominal major diameter of the thread. The other blade holder shall have a head on one end, except that for type II frames, the front end of the blade may be held by a pin secured in the frame. A friction washer shall be provided between the thumb nut and the frame or sleeve. If the threaded holder is located at the rear, sufficient clearance between the handle and holder shall be provided to preclude the operator's hand or fingers from contacting the holder under normal service conditions.

3.4 Blade holders.—Blade holders shall hold the blade rigidly and in true alignment with the centerline of the frame. Blade bearing surfaces shall be smoothly finished to provide a flat bearing surface for the blade. Blade holders for types I, II, and III frames, except type I, class 1, style A, grade b, shall be of such design that will preclude the blade from rotating more than a total of 2 degrees or 1 degree in either direction when subjected to the test specified in 4.4.2.

3.5 Pins and rivets.—Each blade-holding device of types I, II, III, and type IV, class 1 shall be fitted with a pin approximately ⅛ inch (0.124 to 0.130 inch) in diameter, driven through and rigidly secured, or the pin may be an integral part of the blade-holding device. The pin shall be inclined at a slight angle so as to force the blade against the blade-holding device when tightened. All rivets shall be of high-quality steel.

3.6 Identification marking. — Each frame shall be marked in a legible and permanent manner with the manufacturer's name or with a trade-mark of such known character that the source of manufacture may be readily determined.

3.7 Finish.

3.7.1 *Surface.*—All frames and parts shall be smoothly finished.

3.7.2 *Coatings.* — Unless otherwise specified, material subject to corrosion or rust shall have one or a combination of the following coatings (see 6.2):

3.7.2.1 *Bright chromium plate.*—The coating shall be electrodeposited nickel, followed by chromium, the minimum thickness being 0.0002 inch for nickel and 0.00001 inch for chromium.

3.7.2.2 *Nickel plate.*—The coating shall be electrodeposited nickel, and shall be not less than 0.0002 inch thick.

3.7.2.3 *Cadmium plate.*—The coating shall be electrodeposited cadmium not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.7.2.4 *Zinc plate.*—The coating shall be electrodeposited zinc not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

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3.7.2.5 *Hot phosphating.* — The coating shall consist of a heavy chemically produced phosphate coating followed with a coating of corrosion-preventive compound.

3.7.2.6 *Oxide coating.*—The coating shall consist of a heavy chemically produced oxide coating followed with a coating of corrosion-preventive compound.

3.8 Threads.—All threaded parts shall be in accordance with National Bureau of Standards Handbook H28 and the latest supplement thereto.

3.9 Repair parts.—The successful bidders supplying frames covered by this specification shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances and clearances, in order that such parts may be completely interchangeable, replaced, or adjusted without requiring modification.

3.10 Type I, hack, adjustable length.

3.10.1 *Class 1, style A, grades a and b, and style B, flat frame, open grip, shallow and medium throat.*—Class 1 frames shall have open pistol-grip type handles in accordance with 3.10.1.2 and 3.10.1.3. Blade-holding devices shall be adjustable to any one of four positions, 90 degrees apart, and means shall be provided for retaining them in these positions when the blade is tightened. The frame shall have the thumb-nut tension style blade-holding design in accordance with 3.3.2.2, conform to table I, and shall be similar to figure 1.

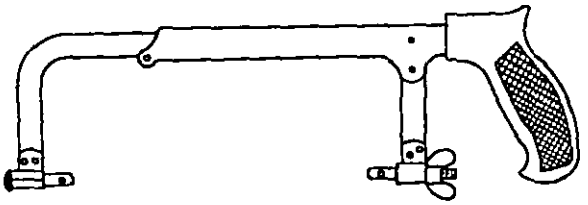


FIGURE 1.—Type I, class 1, styles A and B, hack, adjustable length, flat frame, open grip, shallow and medium throat.

3.10.1.1 *Grades a and b.*—Grades a and b frames shall conform to all requirements specified, except that grade b is not required to be subjected to the torsion test of 4.4.2.

3.10.1.2 *Handles.* — Handles shall be of either pressed sheet steel, malleable iron, or a composition material such as hard rubber or plastic. Sheet-steel handles shall be of material not less than 0.037 inch thick before forming. The handle-grip section of all handles shall be knurled or impressed to provide a secure gripping surface, and set at an angle of not less than 25 degrees nor more than 35 degrees relative to a perpendicular line drawn through the longitudinal axis of the frame.

TABLE I.—*Dimensions, weight, and tension load for type I, class 1, style A, grades a and b, and style B frames*

Style	Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
		Minimum	Maximum		Minimum
	Inches	*Inches*	*Inches*	*Pounds*	*Pounds*
A	10, 12	2½	2 31/32	1⅛	225
B	10, 12	3	3⅞	1½	275

NOTE.—Frames of 8- to 12-inch blade capacity will be acceptable provided they conform to all other requirements herein.

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3.10.1.3 *Handle construction.*—When handles are of sheet steel, the entire length of seams and edges of the handle-grip section shall have a continuous weld. The weld shall be finished smooth (see 3.3.1). Sheet-steel handles that are spotwelded, tackwelded, or have rivets, screws, or pins in the handle-grip section will not be acceptable. Sheet-steel and malleable-iron handles shall be securely attached to the frame by not less than two rivets of at least ⅛-inch diameter; however, the rivets shall not be within the confines of the handle-grip section. Composition handles shall be supported by a steel shank integral with or securely fastened to the frame by not less than two rivets of at least ⅛-inch diameter. The steel shank shall extend into the composition material not less than 2½ inches.

3.10.2 *Class 2, flat frame, closed grip.*—Class 2 frames shall conform to 3.9.1, except the handle shall be closed pistol-grip type. If composition material is used for the handle grip, it shall be either molded around the metal or securely fastened to preclude loosening during service. To assure a comfortable handgrip, composition material secured to the metal shall be properly countersunk or recessed to receive the screw or rivet fasteners. Frames shall conform to table II and shall be similar to figure 2.

3.10.3 *Class 3, round frame, closed grip.*—Class 3 frames shall conform to 3.10.1 except as specified herein. The handles shall be closed pistol-grip type of aluminum or steel construction. When of steel construction the handle-grip section shall be enclosed within a molded composition material such as plastic. The frame shall consist essentially of either a solid bar section which is adjustable by sliding into a tubular rear section, or a one-piece tubular section which is adjustable by sliding into a hole of not less than 3 inches long in the handle. When two-piece frames are furnished, the tubular rear section shall be inserted and securely fastened into the handle to a depth of not less than 3 inches. Both frames shall have means provided to conveniently lock the adjustable section securely in position for both 10- and 12-inch blades. The clearance between the solid bar and tubular section of two-piece frames, and the clearance between the tubular section and the hole in one-piece frames shall not exceed 0.004 inch. Class 3 frames shall conform to table III and shall be similar to view A or B of figure 3.

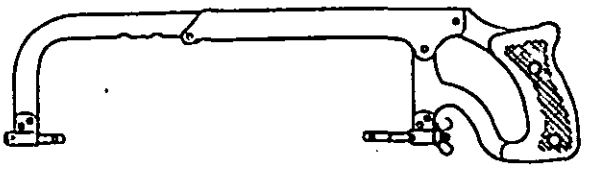


FIGURE 2.—Type I, class 2, hack, adjustable length, flat frame, closed grip.

TABLE II.—*Dimensions, weight, and tension load for type I, class 2 frames*

Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
	Minimum	Maximum		Minimum
Inches	*Inches*	*Inches*	*Pounds*	*Pounds*
10, 12	3	3¼	1¼	275

NOTE.—Frames of 8- to 12-inch blade capacity will be acceptable provided they conform to all other requirements herein.

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TABLE III.—*Dimensions, weight, and tension load for type I, class 3 frames*

Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
	Minimum	Maximum		Minimum
Inches	*Inches*	*Inches*	*Pounds*	*Pounds*
10, 12	3	3¾	1½	300

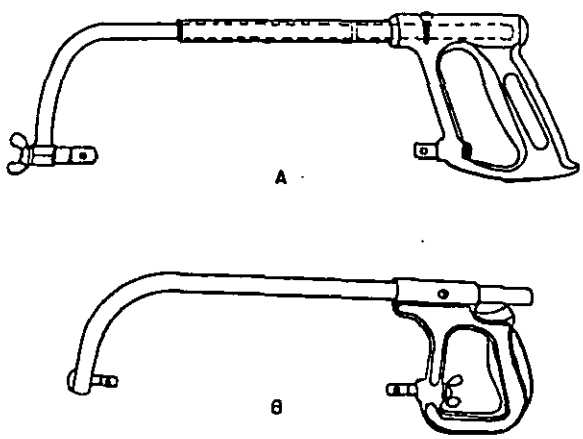


FIGURE 3.—Type I, class 3, hack, adjustable length, round frame, closed grip.

3.11 Type II, hack, adjustable throat. — Type II frames shall have a rear blade-holding device and the thumb-nut-tension style as specified in 3.10.1. The handle shall be of aluminum, have a knuckle guard for protecting the operator's hand, and shall be of sturdy construction. The frame proper shall be of good-quality steel and readily detachable from the handle in order to provide a deep throat or medium throat as desired. The area near each pinhole in the frame back shall be properly tempered or of sufficient thickness to preclude elongation of either hole when subjected to hard service in the deep throat position. They shall conform to table IV and shall be similar to figure 4.

TABLE IV.—*Dimensions, weight, and tension load for type II frames*

Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame				Weight (approximate)	Tension between saw grips	
	In low position		In high position			Minimum	
	Minimum	Maximum	Minimum	Maximum		Low position	High position
Inches	*Inches*	*Inches*	*Inches*	*Inches*	*Pounds*	*Pounds*	*Pounds*
12	2½	3½	6¾	7⅝	1⅞	275	125

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TABLE V.—*Dimensions, weight, and tension load and type III, class 1 frames*

Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
	Minimum	Maximum		Minimum
Inches	*Inches*	*Inches*	*Pounds*	*Pounds*
12	3	3½	1¼	275

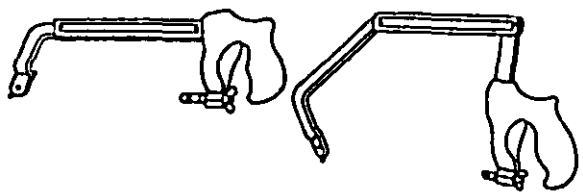


FIGURE 4.—Type II, hack, adjustable throat.

3.12 Type III, hack, nonadjustable.

3.12.1 *Class 1, general purpose.*—Class 1 frames shall conform to 3.10.1, except they shall be nonadjustable, shall conform to table V, and shall be similar to figure 5.

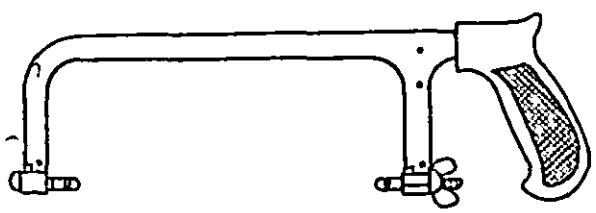


FIGURE 5.—Type III, class 1, hack, nonadjustable, general purpose.

3.12.2 *Class 3, deep throat.* — Class 3 frames shall be of steel and fitted with the conventional style of carpenter's hand-saw handle rigidly secured to the frame by means of at least three telescopic oval head screws having body diameters of not less than 5/16 inch, or equally suitable means. The grip portion of the handle shall set at an angle of not less than 40 degrees nor more than 50 degrees with relation to the vertical portion of the frame to which the handle is secured. Handles shall be of composition material or any suitable tough hardwood, thoroughly seasoned, clear straight grained, and shall be finished smooth and moistureproofed. Blade-holding devices shall be as specified in 3.10.1, and blade tension provided by the thumb-nut-tension style specified in 3.3.2.2. Frames shall conform to table VI and shall be similar to figure 6.

TABLE VI.—*Dimensions, weight, and tension load for type III, class 3 frames*

Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
	Minimum	Maximum		Minimum
Inches	*Inches*	*Inches*	*Pounds*	*Pounds*
12	9¾	10¼	4	200

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TABLE VII.—*Dimensions, weight, and tension load for type IV, class 1 frames*

Capacity (blade length)	Depth from teeth of blade to outside edge or projection of frame (maximum)	Weight (approximate)	Tension between saw grips (minimum)
Inches	*Inches*	*Ounces*	*Pounds*
10	1¼	8	150

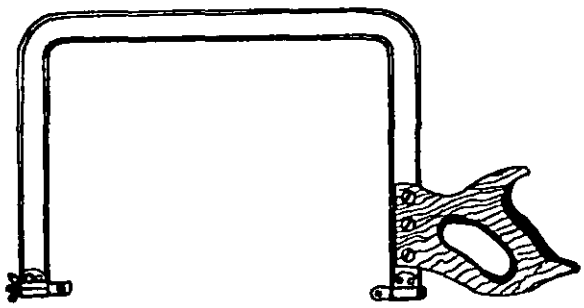


FIGURE 6.—Type III, class 3, hack, nonadjustable, deep throat.

3.13 Type IV, hack, close quarters.

3.13.1 *Class 1, straight frame.*—Class 1 frames shall be of steel and have a straight-style handle in normal alignment with the blade. The handle shall be of metal, composition material, or any suitable tough hardwood, thoroughly seasoned, clear straight grained, and shall be finished smooth and moistureproofed. Tension shall be applied to the blade either by means of the handle-tension style (see 3.3.2.1), or the thumb-nut-tension style (see 3.3.2.2) located on the forward saw grip. They shall conform to table VII and shall be similar to figure 7.

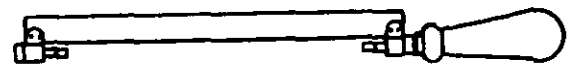


FIGURE 7.—Type IV, class 1, hack, close quarters, straight frame.

3.13.2 *Class 2, triangular frame.*—Class 2 frames shall be of heat-treated steel rod suitably shaped to provide sufficient spring tension to hold the blade rigidly and in true alignment during normal use. The ends of the rod shall be slotted and notched to hold a pin-end type of saw blade. The frame and handle shall be of one-piece construction, conforming to table VIII and shall be similar to figure 8.

TABLE VIII.—*Dimensions, weight, and tension load for type IV, class 2 frames*

Capacity (blade length)	Depth from teeth of blade to outside edge of frame at handle (maximum)	Weight (approximate)	Tension between saw grips (minimum)
Inches	*Inches*	*Ounces*	*Pounds*
6	2¼	4	35

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TABLE IX.—*Dimensions, weight, and tension load for type V, style A frames*

Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
	Minimum	Maximum		Minimum
Inches	*Inches*	*Inches*	*Ounces*	*Pounds*
3 to 6, incl.	2	3	6	25

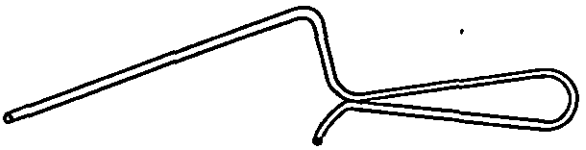


FIGURE 8.—Type IV, class 2, hack, close quarters, triangular frame.

3.14 Type V, jeweler's, adjustable length.

3.14.1 *Style A, medium throat.*—Style A frames shall be of steel and shall have handles of composition material, or any suitable hardwood thoroughly seasoned, straight clear grained, finished smooth, and lacquered or suitably painted. Blade tension shall be provided by the handle-tension style specified in 3.3.2.1 or by the thumb-nut-tension style specified in 3.3.2.2. When specified (see 6.2), frames shall be furnished with blade-holding devices that are rotatable around 360 degrees and locked in any position. Frames shall conform to table IX and shall be similar to figure 9.



FIGURE 9.—Type V, style A, jeweler's, adjustable length, medium throat.

3.14.2 *Style B, deep throat.* — Style B frames shall be of steel and shall have handles as specified in 3.14.1. The frames shall provide sufficient spring tension to hold the blade rigidly during normal use without requiring blade-tension adjustment. They shall conform to table X and shall be similar to figure 10.

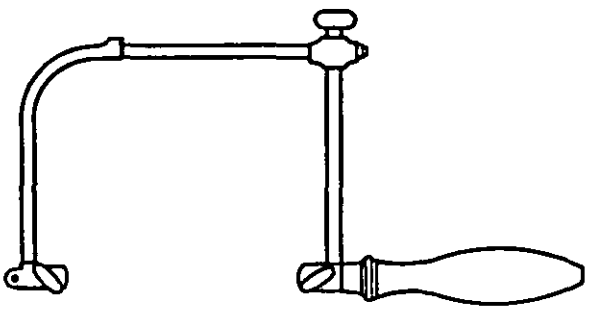


FIGURE 10.—Type V, style B, jeweler's, adjustable length, deep throat.

TABLE X.—*Dimensions, weight, and tension load for type V, style B frames*

Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
	Minimum	Maximum		Minimum
Inches	*Inches*	*Inches*	*Ounces*	*Pounds*
3 to 6, incl.	5¼	6¼	8	20

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TABLE XI.—*Dimensions, weight, and tension load for type VI, styles A and B frames*

Style	Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
		Minimum	Maximum		Minimum
	Inches	*Inches*	*Inches*	*Ounces*	*Pounds*
A	5	2¼	3¼	4	25
B	5	4¾	5¼	6	15

3.15 Type VI, jeweler's, nonadjustable.

3.15.1 *Style A, medium throat, and style B, deep throat.*—Styles A and B frames shall be of steel and have handles as specified in 3.14.1. Blade tension shall be provided by either or both the handle-tension style and the thumb-nut-tension style specified in 3.3.2.1 and 3.3.2.2, respectively. The blade-holding devices shall be rotatable around 360 degrees and locked in any position. Frames shall conform to table XI and shall be similar to figure 11.

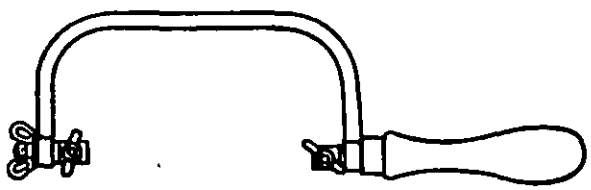


FIGURE 11.—Type VI, styles A and B, jeweler's nonadjustable, medium and deep throat.

3.16 Type VII, coping.

3.16.1 *Style A, medium throat, and style B, deep throat.*—Styles A and B frames shall be of steel and have handles as specified in 3.14.1. Blade tension shall be provided by the handle-tension style specified in 3.3.2.1. Blade-holding devices shall be designed for holding pin-end blades, shall be rotatable around 360 degrees, and means shall be provided to facilitate their turning. They shall conform to table XII and shall be similar to figure 12.

3.17 Workmanship. — The workmanship shall be first class throughout. Frames shall be free from rust, fins, burrs, external sharp or rough edges, corners or surfaces and defects which may impair their serviceability or durability.

TABLE XII.—*Dimensions, weight, and tension load for type VII, styles A and B frames*

Style	Capacity (blade length)	Depth from teeth of blade to nearest inside edge or projection of frame		Weight (approximate)	Tension between saw grips
		Minimum	Maximum		Minimum
	Inches	*Inches*	*Inches*	*Ounces*	*Pounds*
A	6½	4½	5 1/16	7	50
B	6½	5 1/16	6⅞	9	40

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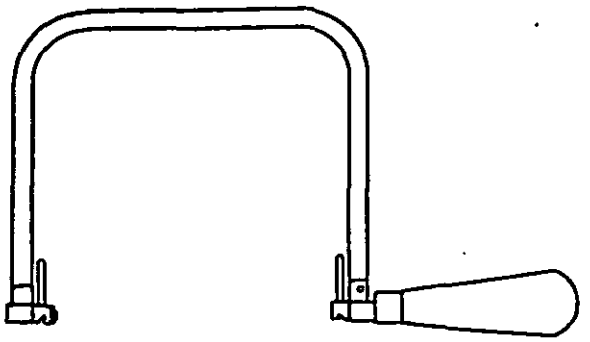


FIGURE 12.—Type VII, styles A and B, coping, medium throat and deep throat.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance inspection.

4.1.1 *Lot.*—For purposes of sampling, a lot shall consist of frames of one type, class style and grade offered for delivery at one time.

4.1.2 *Sampling for visual and dimensional examination.*—A random sample of frames shall be selected from each inspection lot of material offered for Government examination of visual and dimensional characteristics in accordance with Military Standard MIL–STD–105 at Inspection Level II for lots containing 110 items or less and Inspection Level I for lots containing more than 110 items. The Acceptable Quality Level shall be 1.5 percent defective.

4.1.3 *Sampling for lot acceptance tests except flammability test (see 4.4.3.3).*—From each lot of material offered for Government tests a random sample of frames shall be selected in accordance with the appendix to Military Standard MIL–STD–105 at Inspection Level L–7. The Acceptable Quality Level shall be 2.5 percent defective. However, the sample size shall be the number associated with the letter in the applicable table of the appendix and the acceptance number shall be zero until the acceptance number for the Acceptable Quality Level is one or greater.

4.1.4 *Sampling for flammability test (if applicable) (see 4.4.3.3).*—A random sample of frames shall be selected from each lot of material offered for Government testing in accordance with the appendix to Military Standard MIL–STD–105 at Inspection Level L–3. If any sample frame fails the test the lot shall be rejected.

4.2 Visual and dimensional examination.—Each of the sample frames selected in accordance with 4.1.2 shall be visually and dimensionally examined to verify compliance with this specification. Any frame in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective frames in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming frames.

4.3 Lot acceptance test except flammability.—Each of the sample frames selected in accordance with 4.1.3 shall be tested in accordance with 4.4 except 4.4.3.3 to verify compliance with the requirements of this specification. Any sample frame which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming frames in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Application of tension load between saw grips.* — Shallow center punch marks shall be applied on both ends of the frame, preferably just above the blade-holding devices. The distance between these two punch marks shall be measured while the frame is subjected to the following tension loads:

(a) For types I, II, III, and type IV, class 1 frames, a tension load of 20 pounds is applied.

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(*b*) For type IV, class 2 and type VII frames, a tension load of 10 pounds is applied.

(*c*) For types V and VI frames, a tension load of 5 pounds is applied.

4.4.1.1 Contraction of the frame shall be not more than ¼ inch when the tension is increased to the number of pounds specified in the applicable table for the frame being tested, except that 5/16-inch will be allowed on type III, class 3 frames, and ⅝-inch will be allowed on type VII frames. There shall be no permanent set when the load is decreased to the applicable tension load specified in (a), (b), or (c) as indicated by a difference in the distance between the punch marks. If the frame shows a permanent set of less than 1/16 inch, the test shall be repeated once. After the second test, if the frame shows no further permanent set, it will be considered to have met the test satisfactorily. Adjustable frames shall be set to maximum blade capacity when subjected to this test, and care shall be exercised to see that the loads are applied to the saw grips.

Note.—The apparatus shown on figure 13 may be used, and is approved for conducting the tension test on types I, II, and III frames. With suitable adapters and additional holes (see figure 13, holes C1, C2, C3, and C4) types IV, V, VI, and VII frames may be tested similarly. A small loop having a single end projection of 90 degrees may be formed from ⅜-inch-diameter steel and would serve as a simple adapter in most cases. It is not intended to restrict the test apparatus to that described; therefore, other suitable methods may be employed.

4.4.1.2 The apparatus shown on figure 13 consists chiefly of an angle iron (A), a notched bar (B), and a weight (D). The angle iron (A) and bar (B) are held together at (C) by a pin. The holes (E–E) are 12 inches apart (when bar (B) is in a horizontal position) and are directly in line with each other. The distance between (C) and (E) is two inches. The tension load between saw grips is computed by multiplying the force in pounds by one-half the arm length in inches (50-pound load applied 10 inches from (C) equals 250 pounds, and so forth). The arm is

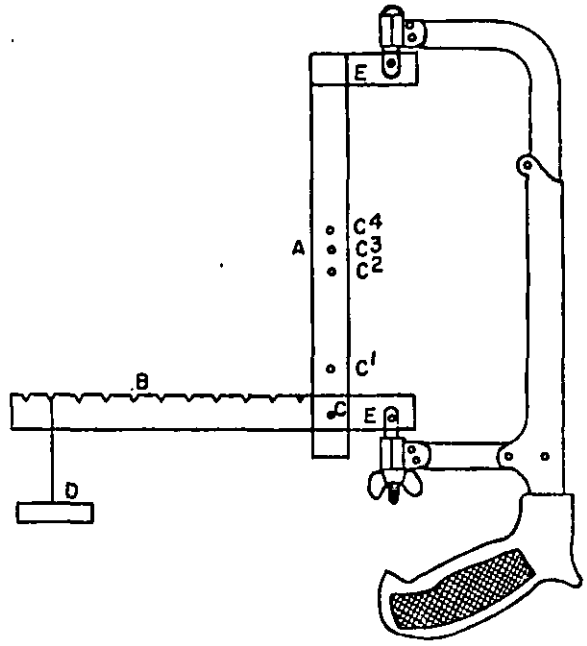


FIGURE 13.—Apparatus for applying tension load test.

the distance from (C) to the weight attached on the bar (B). The method of applying the test load is as follows: Grasp angle iron (A) in a vertical position, in an ordinary bench vise. Place frame in position shown with grip pins of frame in holes (E–E). By tightening up on the thumb nut or handle of frame the load applied thereon is balanced. When the bar (B) is horizontal the load applied thereon is a direct measurement of the tension between the grips of the hacksaw frame. Care shall be exercised in loading of the bar as the true tension cannot be determined when the bar sags below the horizontal plane. C1, C2, C3, and C4 holes may be suitably located and used when testing frames with blade length capacities of 5, 6, 6½, and 10 inches.

4.4.2 *Torsion test.*—While types I, II, and III frames, except type I, class 1, style A, grade b, are under a tension load of approximately 100 pounds, the blade holders shall retain the position to which they may be set without damage to any part, nor shall they rotate more than a total of 2 degrees (1 degree in either direction) when a torsional force of 35 inch-pounds is applied to either of

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the blade holders. For type II frames, this applies only to the rear blade holder.

4.4.3 *Composition-material handles.*

4.4.3.1 *Impact test.*—The frames shall be held handle downward and dropped on a concrete floor or slab from a height of 10 feet. After impact, there shall be no flaking off or breakage of the handle.

4.4.3.2 *Heat-distortion test.*—The handle of the frame shall be completely immersed in water heated to approximately 160° F. for a period of 30 minutes. At the end of this period, the sample shall be removed from the water and shall show no significant signs of distortion or blistering.

4.4.3.3 *Flammability test.*—The frame shall be clamped in a support with the handle extended outward in a horizontal position. Under the handle there shall be clamped a piece of 10-mesh Bunsen gauze about 5 inches square, in a horizontal position ¼ inch below the bottom of the handle and with about ½ inch of the handle extending beyond the edge of the gauze. A Bunsen burner with an efficient flame ½ to ¾ inch in height, shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the handle. At the end of 30 seconds, the flame shall be removed and the specimen allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the handle. In case the handle does not continue to burn after the first ignition, the burner shall be placed under the free end for a period of 30 seconds immediately following extinction of the flame. If the handle does not continue to burn until the flame has reached the frame end of the handle after the second ignition, the sample shall be considered self-extinguishing. When tested as described, the handles shall not burn at a rate exceeding 2 inches per minute.

4.5 *Rejected lots.*—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconfoming frames. Samples shall again be selected from such resubmitted lots and examined to verify compliance with the specification requirements.

5. PREPARATION FOR DELIVERY

(Nonmilitary agencies. — Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.

5.1.1 *Level A.*—Frames shall be preserved and packaged in accordance with Level A in Military Specification MIL–P–15424.

5.1.2 *Level C.*—Frames shall be preserved and packaged in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Frames shall be packed in accordance with Level A in Military Specification MIL–P–15424.

5.2.2 *Level B.*—Frames shall be packed in accordance with Level B in Military Specification MIL–P–15424.

5.2.3 *Level C.* — Frames preserved and packaged in accordance with 5.1.1 or 5.1.2 as specified (see 6.2) shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Consolidated Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military.*—In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Militay Standard MIL–STD–129.

5.3.2 *Nonmilitary.*—Marking for shipment shall be in accordance with the standard marking of Federal Specification PPP–P–31.

6. NOTES

6.1 Intended use.

6.1.1 Type I, classes 1, 2, 3 and type III, class 1 frames are primarily for general shop

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use in cutting ferrous and nonferrous metals. Relative to type I, class 1, style A frames, grade a is intended to withstand harder usage than grade b and be more rigid to facilitate guiding straight cuts.

6.1.2 Type II frames are primarily for intermittent use in deep cutting metal plates, pipes, etc., and alleviate the need of carrying both a deep-throat and ordinary-throat hacksaw on normal maintenance jobs.

6.1.3 Type III, class 3 frames are primarily for use in deep cutting channel iron, I-beams, or girders.

6.1.4 Type IV, class 1 frames are primarily for use in cutting miscellaneous metals in very close quarters.

6.1.5 Type IV, class 2 frames are primarily used for light-duty cutting of metal, plastic, or wood, in medium close quarters.

6.1.6 Types V and VI frames are primarily used for very light duty in cutting precious and common metals by jewelers, instrument markers, etc.

6.1.7 Type VII frames are primarily used for light-duty cutting of wood or soft miscellaneous material.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein and should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, style, and grade required if applicable. (See 1.2.)
(*c*) Specify if a particular type of coating is required. (See 3.7.)
(*d*) Blade-holding devices rotatable 360 degrees, if required for type V, style A jeweler's frame. (See 3.14.1.)
(*e*) Whether preservation and packaging should be level A or C, and whether packing should be by level A, B, or C (see section 5).

6.3 Transportation description.—The transportation description applicable to these items is:

Tools, not otherwise indexed by name.
Carload weight 30,000 pounds.
Mechanics' hand tools, not otherwise indexed.
Truckload weight 30,000 pounds.

6.4 This specification conforms with Simplified Practice Recommendation R90–49 insofar as blade capacity for hacksaw frames is concerned.

6.5 Superseding data.—In this specification:

Type A of Federal Specification GGG–F–671 has been deleted.
Type B of Federal Specification GGG–F–671 shall be known as type I, class 1, styles A and B.
Type C of Federal Specification GGG–F–671 shall be known as type III, class 3.

6.6 Type III, class 2 of Federal Specification GGG–F–671a nonadjustable, heavy duty hack saw frame has been deleted by this revision because of the lack of Government requirements for this class of heavy duty frame.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—O E Q Sig T
Navy—Sh A MC Or Y
Air Force.

Copies of this specification may be purchased for 10 cents each.

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